

PANCaKES: Panoptic Adversarial Nuclei Classification and Knowledge-Enhanced Segmentation using GANs

¹Abdullah Ibrahim ¹Ahmed Abdullah
1227552@lhr.nu.edu.pk 1227503@lhr.nu.edu.pk
¹Noman Jaffar
1227535@lhr.nu.edu.pk

¹FAST-NUCES, Department of Data Science & AI

1 Survey

Q1. Which LLM do you generally use to get coding help?

- ChatGPT (GPT-4, GPT-3.5)

Q2. How comfortable are you with Copilot?

- Comfortable

Q3. How frequently did you use Copilot for this phase?

- Occasionally (a few times)

Q4. How do you feel about using Copilot for basic tasks and gradually increasing project complexity?

- Neutral

2 Problem Statement

In the rapidly evolving field of medical image analysis, histopathological imaging stands at the forefront, offering unparalleled insights into disease diagnosis and treatment planning [1] [2]. Our model represents a significant improvement over nucleus segmentation in histopathology and is capable of negating significant defects that are present in existing systems.

Medical image repositories such as PanNuke tend to suffer from rampant issues such as overlapping nuclei, noisy data, and class imbalance. Such issues negatively affect baseline model performance. Without our model, clinical practice and research would suffer significantly. This underscores the importance of our model and suggests the repercussions of our model not being there.

2.1 Improved Nucleas Segmentation

Implication: Proper nuclei segregation is important for accurate diagnosis of the majority of diseases, including cancer. Our model demonstrates better accuracy on hard cases of overlapping and eccentric nuclei and unbalanced datasets, in which other models could prove useless. Segmentation errors result in misdiagnosis, incorrect treatment regimens, and overall diminished quality of care.

2.2 Class Imbalance Handling Why It Matters?

Class imbalance is prevalent in histopathological images where rare but salient classes of nuclei are overshadowed by more common classes. We address this with the most powerful methods that involve synthetic data augmentation, data augmentation using GAN, and regularization.

2.3 Strong Performance on Noisy Images

Relevance: Medical images tend to have noise due to inconsistency in imaging or staining procedures. The model used in this research uses the best denoising and segmentation algorithms present, with the aim of being invariant when presented with noisy input. Without our model, other models would be prone to noise, resulting in erroneous segmentation and reduced accuracy in AI diagnoses.

2.4 Generalizability and Scalability Significance of the Study

Our model is intended to be extremely generalizable across datasets with the help of operations like cross-validation, data augmentation, and SAM v2 architecture. Such an ability guarantees its applicability to other real-world health care settings. Other models will not generalize across datasets, and thus they would be less valuable and would be incapable of delivering AI as a viable alternative in the medical imaging arena as a whole.

2.5 Driving Medical Research and Innovation Why It Matters?

Our model enables tumor classification, disease grading, and survival prediction research by making feasible precise nuclear segmentation. Medical AI research is accelerated with the precise segmentation masks generated by our model.

2.6 Clinical Risk Management

Precise segmentation lessens the danger of diagnostic mistake, with immediate implications for patient safety. Our model enhances diagnosis accuracy and enables better treatment options.

The given model is not another gadget, but a jump of cosmic proportions for medical image analysis. Without it, lingering issues would remain to exert a stranglehold on research, diagnosis, and patient treatment. Providing accurate,

scalable, and noise-resilient segmentation, the model completes critical gaps and sets the stage for enhanced AI diagnosis and medical breakthroughs.

3 Dataset Description and Source

The data is sourced from the Huggingface dataset `RationAI/PanNuke`. The PanNuke dataset, a comprehensive collection of annotated histopathological images, serves as a pivotal resource in this domain [3]. Comprising 8,000 images with approximately 100,000 annotated nuclei across various tissue types, it provides a rich foundation for developing advanced image processing techniques [3].

3.1 Columns Information

The dataset is organized into three folds: `fold1`, `fold2`, and `fold3`, consistent with the original dataset structure.

Each fold contains data in a tabular format with the following four columns:

- **Image:** The RGB tile of the sample.
- **Instances:** A list of nuclei instances. Each instance represents exactly one nucleus and is in binary format (1 - nucleus, 0 - background).
- **Categories:** An integer class label for each nucleus, corresponding to one of the following categories:
 1. Neoplastic
 2. Inflammatory
 3. Connective
 4. Dead (*Severely Underrepresented*)
 5. Epithelial
- **Tissue:** The integer tissue type from which the sample originates, belonging to one of the following categories:

– Adrenal Gland	– Head & Neck	– Skin
– Bile Duct	– Kidney	– Stomach
– Bladder	– Liver	– Testis
– Breast	– Lung	– Thyroid
– Cervix	– Ovarian	– Uterus
– Colon	– Pancreatic	
– Esophagus	– Prostate	

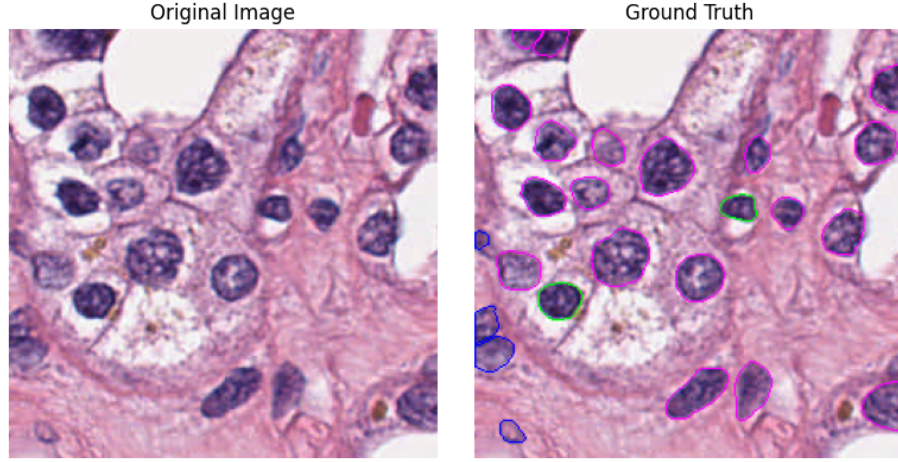


Figure 1: Comparison of an original histopathology image (left) and its corresponding ground truth annotations (right) from the dataset.

3.2 Challenges with Dataset

The dataset presents challenges such as class imbalance and inherent noise in nuclear structures, which can hinder the performance of traditional deep learning models. To address these issues, we propose a multi-step pipeline that integrates advanced data augmentation, generative modeling, and state-of-the-art segmentation techniques. Our goal is to enhance the dataset’s quality and diversity, thereby improving segmentation accuracy and contributing to more robust medical imaging applications [3] [4].

4 Data Pre-processing

4.1 Meta Data Pre-processing

The comma separated file (csv) file has been imported with some images’ meta information, including cell categories, tissue type, etc. These were stored in the metadata as lists of categories (as strings). These had to be read out as strings and converted into Python list objects.

4.2 Updated File Structure for Data Loader

For each nucleus instance, separate sparse images of the mask were maintained, storage and processing were therefore complex. To simplify, we now have a single image per category where the masks are compressed. Different types of nuclei will no longer have separate images; all instances of the same category will now exist in a single mask, thus enhancing efficiency in data handling.

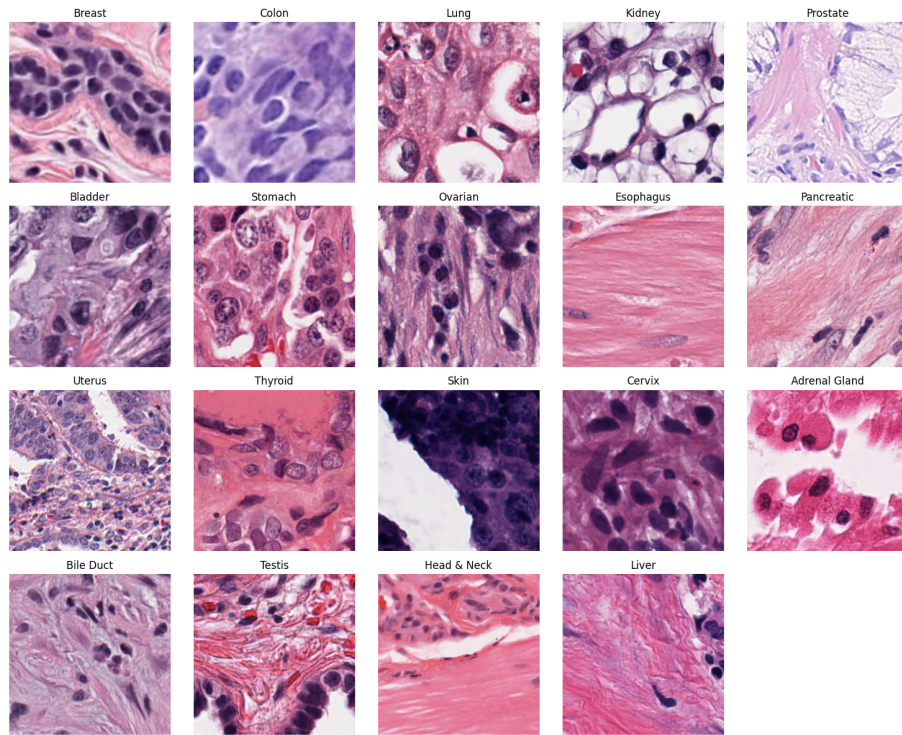


Figure 2: Overview of different tissue types in the dataset, including Breast, Colon, Lung, Kidney, Prostate, and more.

```

--- Fold1
----- image1_path:(image, compressed mask for Neoplastic cells)
----- image1_path:(image, compressed mask for Inflammatory cells)
----- image1_path:(image, compressed mask for Connective cells)
----- image1_path:(image, compressed mask for Dead cells)
----- image1_path:(image, compressed mask for Epithelial cells)
----- image2_path:(image, compressed mask for Neoplastic cells)
-----

--- Fold2
----- image1_path:(image, compressed mask for Neoplastic cells)
----- image1_path:(image, compressed mask for Inflammatory cells)
-----

```

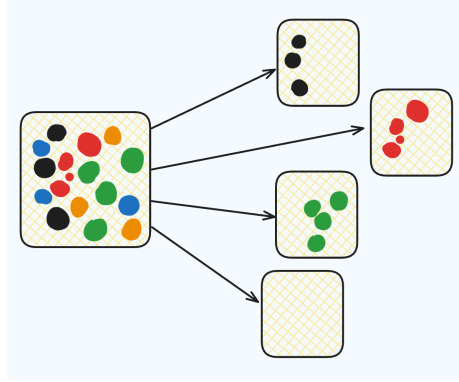


Figure 3: Panoptic Segmentation Data Preparation with Compressed Masks

4.3 Mask Consolidation Using Bitwise Operations

A bitmask was applied individually to every sample sliced from the original image. In order to use present binary masks and mask images, it is necessary to combine their compression and expansion into one mask that would enclose all masked areas that several cell nuclei could cover. We used bitwise operations to combine them into one mask (representing all masked areas).

5 Generative Adversarial Networks Augmentation Pipeline

We began our augmentation pipeline by training a UNet-based model using a custom architecture with attention-enhanced encoder and decoder paths. The architecture is defined using the following configuration:

- **Input size:** 3 channels (RGB)
- **Output size:** 3 channels
- **Resolution:** Configurable via `config.image_size`
- **Layers per block:** 2
- **Block output channels:** (128, 128, 256, 256, 512, 512)
- **Down blocks:**
 - 4 regular ResNet downsampling blocks (`DownBlock2D`)
 - 1 attention-based downsampling block (`AttnDownBlock2D`)
 - 1 additional downsampling block (`DownBlock2D`)
- **Up blocks:**
 - 1 regular upsampling block (`UpBlock2D`)

- 1 attention-based upsampling block (`AttnUpBlock2D`)
- 4 regular upsampling blocks (`UpBlock2D`)

This model was trained to generate synthetic nuclei images from a noise vector. Below we show the initial output of the base model:

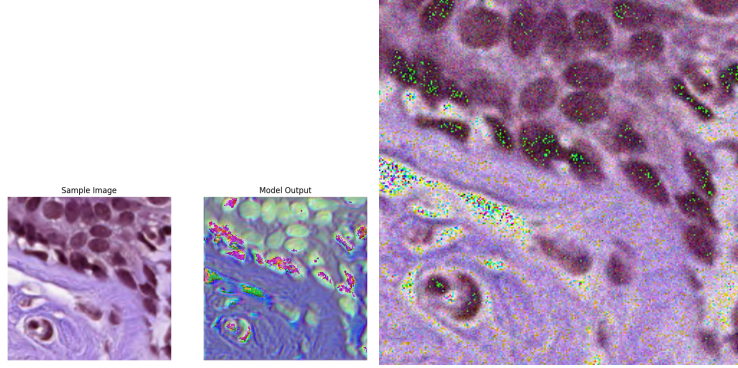


Figure 4: Left: Base model prediction after initialization (Epoch 0), Right: Corresponding input tensor (noise)

Over time, as the UNet was trained, the generated outputs improved drastically in realism. The following figures show the evolution of the generated output from epoch 50 to 300:

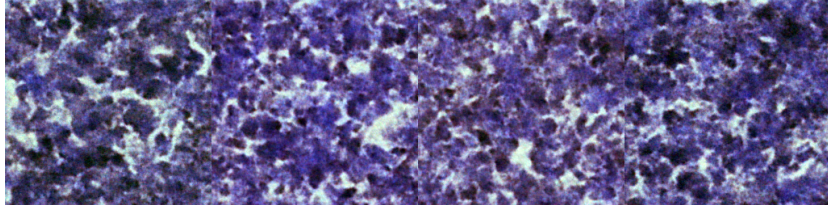


Figure 5: Predicted output at epoch 50

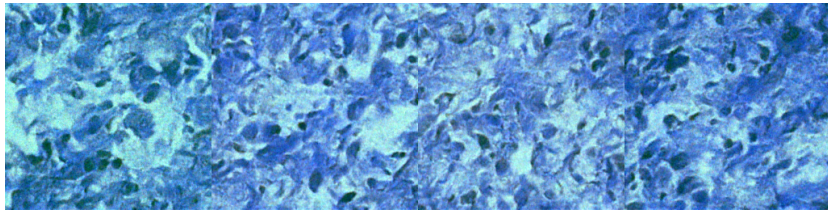


Figure 6: Predicted output at epoch 100

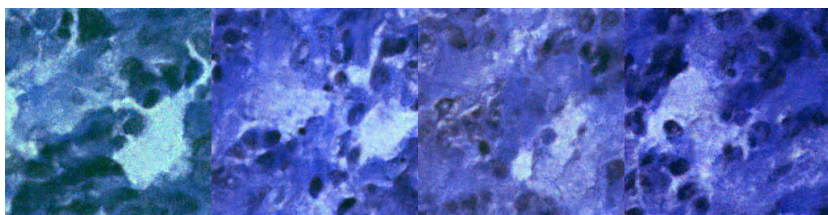


Figure 7: Predicted output at epoch 150

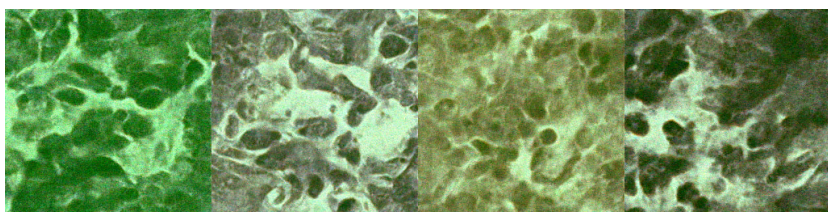


Figure 8: Predicted output at epoch 200

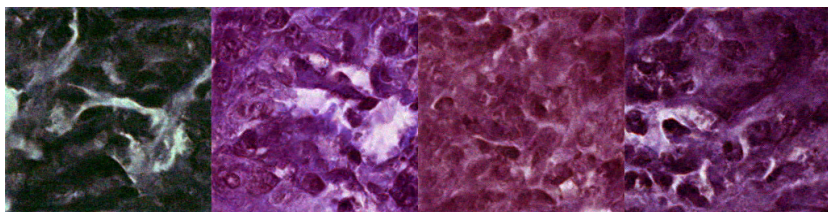


Figure 9: Predicted output at epoch 250

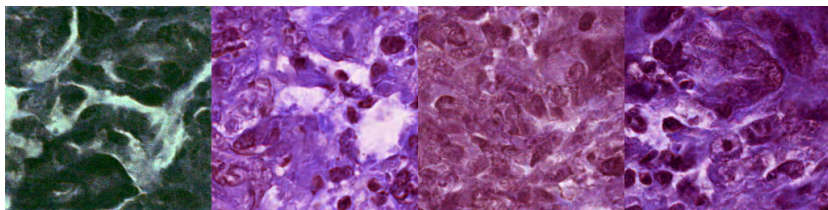


Figure 10: Predicted output at epoch 300

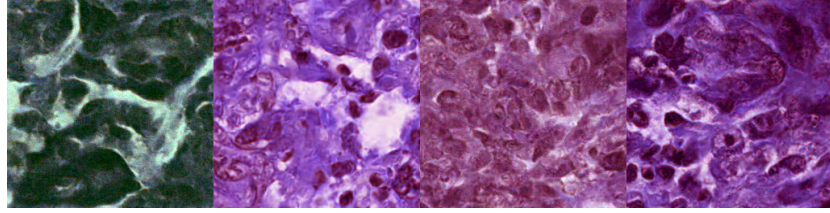


Figure 11: Final predicted output at epoch 300, demonstrating high fidelity to real nuclei structures.

After the training was complete, the model was used to generate synthetic data:

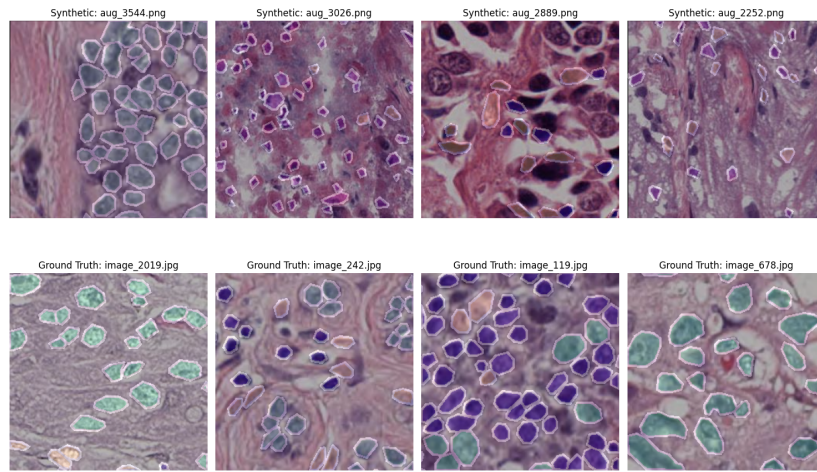


Figure 12: Comparison of real and synthetic data. Visually, there is no clear distinction, validating the quality of generated samples.

Following this augmentation process, the new class distribution after including synthetic images is shown below:

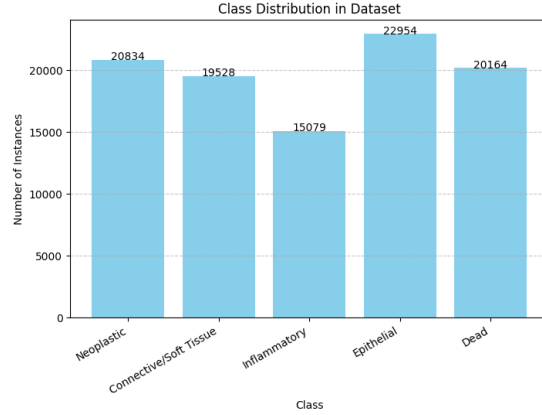


Figure 13: Class distribution after augmentation. While the distribution has improved, perfect balance is not achievable due to inherent variations in cell counts across images.

6 Exploratory Data Analysis

6.1 Cell Type Distribution

The different cell types were analyzed to assess their composition within the dataset. The bar chart on the left indicates the count of each cell type, showcasing that neoplastic and connective cells largely prevail, whereas dead cells are significantly underrepresented. The pie chart on the right provides information regarding the proportion of images containing each specific cell type, thus emphasizing the imbalance within the dataset.

6.2 Tissue Type Distribution

The dataset contains the majority of samples under two tissue types: **Breast** and **Colon**, while other tissues such as **Kidney**, **Bladder**, and **Stomach** have far fewer samples. This discrepancy may impair model performance and must be addressed with approaches like data augmentation or weighted loss functions to provide an equal opportunity for learning across tissue types.

6.3 KDE Plot for Image Intensity Distribution by Cell Type

The kernel density estimation (KDE) plot below illustrates the distribution of mean image intensity across different cell types. Each curve represents a smoothed probability density function for a specific cell type, highlighting the most common intensity values observed. **Dead Cells** exhibit a peak at lower intensity values, whereas **Epithelial** and **Neoplastic Cells** display higher intensity distributions. These patterns help in understanding variations in brightness and contrast among different cellular structures.

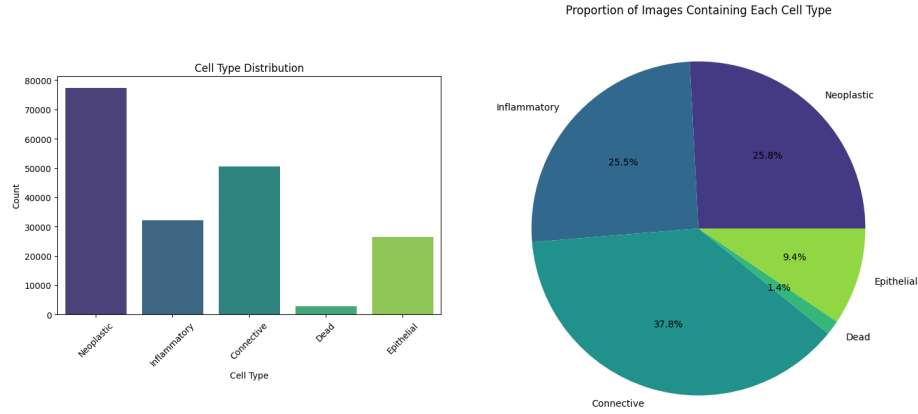


Figure 14: Left: Cell Type Distribution by Count. Right: Proportion of Images Containing Each Cell Type.

6.4 Scatter Plots for Mask Coverage vs. Mean Image Intensity

The scatter plots below represent the relationship between **mask coverage** and **mean image intensity** for different cell types. Each point corresponds to an individual observation, showing how the percentage of image coverage varies with pixel intensity. **Neoplastic and Inflammatory Cells** exhibit wider distributions, suggesting high variability in their intensity and coverage. **Dead Cells** appear less frequent and are more concentrated in specific intensity ranges. These insights are valuable in segmenting and identifying cell types effectively.

6.5 Cell Type Distribution Across Tissues

The heatmap below indicates the distribution of various tissues according to cell types. Color intensity corresponds to the frequency of each cell type in each tissue. **Neoplastic** and **Epithelial** cells are abundant in tissues such as **Breast** and **Colon**. Dead cells are significantly fewer in all tissues. This visualization serves as a guide for identifying class imbalances and considering possible balancing techniques to improve model performance.

6.6 Box Plot for Image Intensity Distribution by Cell Type

The box plot shown below summarizes the statistical data concerning the mean image intensity for each cell type. The central box shows the interquartile range (IQR), while the median is marked by a line within the box. The line segments extending from the box (the whiskers) indicate the smallest and largest values within 1.5 times the IQR, while individual data points beyond the whiskers represent outliers.

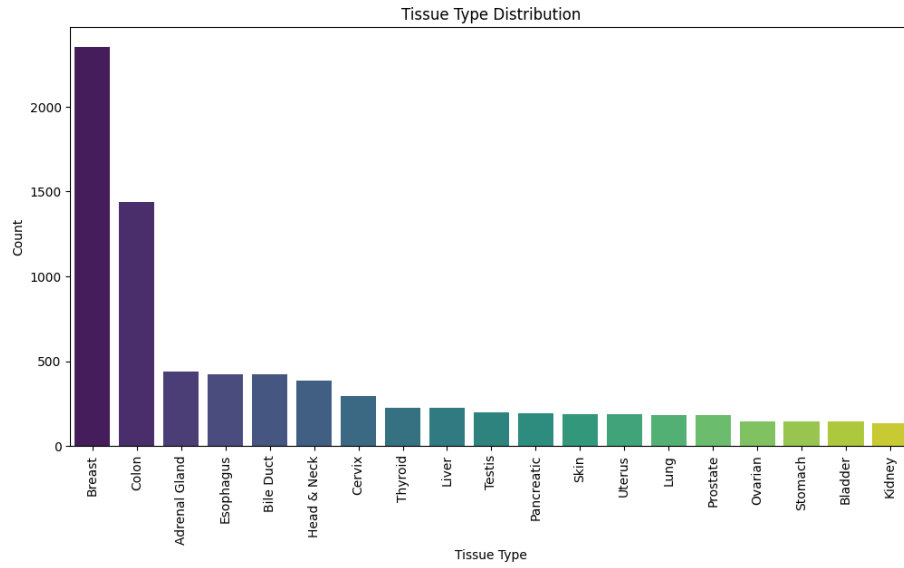


Figure 15: Tissue Type Distribution in the Dataset

- **Dead Cells** exhibit a slightly lower median intensity compared to other cell types, with a more compact distribution.
- **Neoplastic, Inflammatory, Connective, and Epithelial Cells** display similar distributions with high variability, extending from low to high intensity values.

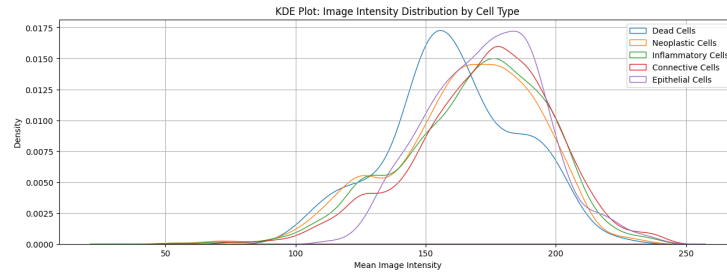


Figure 16: KDE plot of image intensity distribution across different cell types.

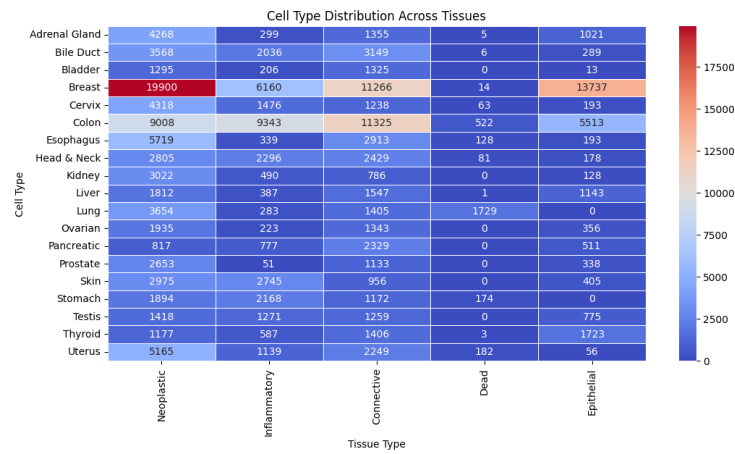


Figure 17: Heatmap of Cell Type Distribution Across Tissues

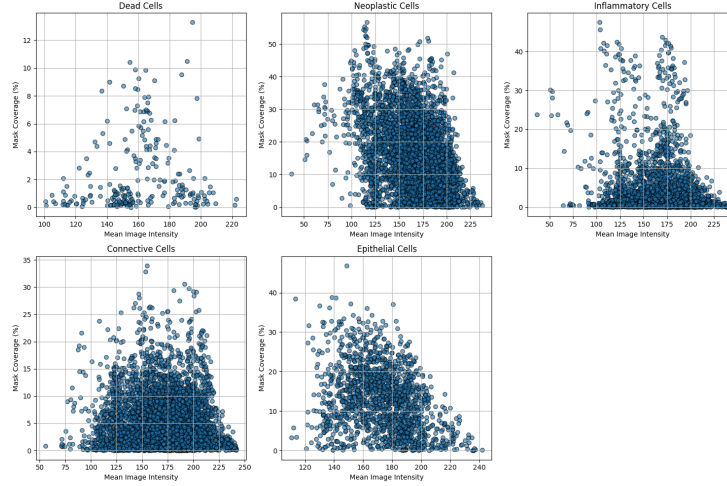


Figure 18: Scatter plots of mask coverage versus mean image intensity for different cell types.

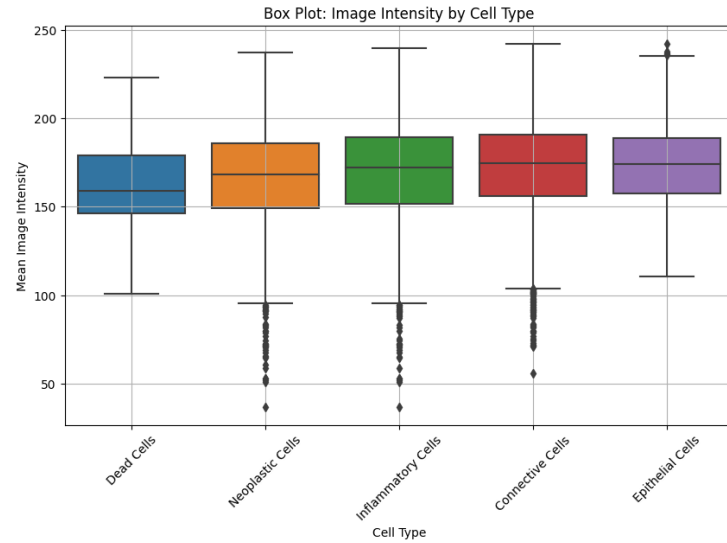


Figure 19: Box plot illustrating the distribution of mean image intensity for different cell types, highlighting variations and outliers.

References

- [1] A. Pedersen, M. Valla, A. M. Bofin, J. P. De Frutos, I. Reinertsen, and E. Smistad, “FastPathology: An Open-Source Platform for Deep Learning-Based Research and Decision Support in Digital Pathology,” *IEEE Access*, vol. 9, pp. 58 216–58 229, 2021.
- [2] “Advancements in medical image analysis,” *ScienceDirect*, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S136184152400269X>
- [3] RationAI, “Pannuke dataset,” Hugging Face, 2023. [Online]. Available: <https://huggingface.co/datasets/RationAI/PanNuke>
- [4] F. Hörst, M. Rempe, L. Heine, C. Seibold, J. Keyl, G. Baldini, S. Ugurel, J. Siveke, B. Grünwald, J. Egger, and J. Kleesiek, “Cellvit: Vision transformers for precise cell segmentation and classification,” *Medical Image Analysis*, vol. 94, p. 103143, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1361841524000689>